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1. INTRODUCTION

1.1 Project Overview

The House Price Analysis project aims to analyze housing market data using Tableau and present meaningful insights through interactive dashboards and visualizations. The project examines various housing attributes such as house prices, house age, renovation status, bedrooms, bathrooms, floors, and sales trends to help users better understand the housing market. Interactive dashboards enable users to explore the data efficiently and identify important patterns that support informed decision-making.

1.2 Purpose

The primary purpose of this project is to transform raw housing data into meaningful visual information using Tableau. The project helps home buyers, property sellers, and real estate analysts understand housing market trends, compare property characteristics, and identify the major factors influencing house prices.

2. IDEATION PHASE

2.1 Problem Statement:

Customer Problem Statement :

The **House Price Analysis** project is designed to understand the challenges faced by home buyers, property sellers, and real estate professionals when analyzing housing market data. This project focuses on identifying the difficulties users encounter while comparing house prices, property features, renovation status, and market trends. By clearly defining these problems, the team aims to develop an effective data visualization solution that simplifies complex housing information and supports better decision-making.

A well-defined customer problem statement helps the team understand user needs and expectations from different perspectives. It enables the identification of important factors affecting house prices and transforms raw housing data into meaningful visual insights. Using **Tableau**, the project presents interactive dashboards and stories that allow users to explore housing trends, compare property characteristics, and make informed decisions based on accurate and easy-to-understand visual analysis.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	I am a home buyer.	Find a house that fits my budget and needs.	I cannot easily compare house prices, locations, and property features.	Housing data is scattered and difficult to understand.	Confused and unsure about making the right buying decision.
PS-2	I am a real estate agent or property analyst.	Analyze housing prices and market trends.	It is difficult to identify price patterns and important factors affecting house values	Large datasets are complex and require proper visualization and analysis.	Time-consuming and less confident in providing accurate recommendations.

			using raw data.		

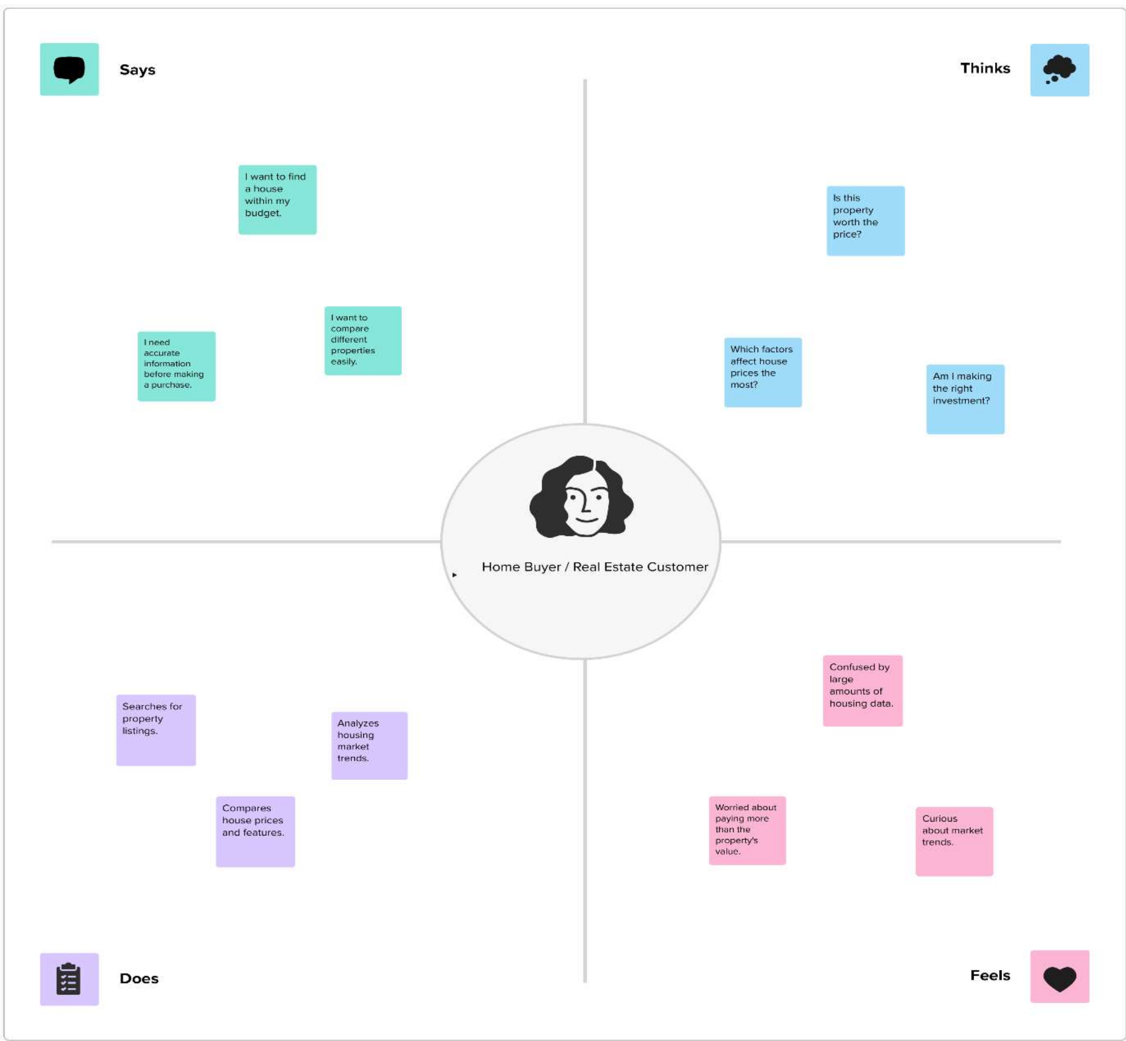
2.2 Empathy Map Canvas:

Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to helps teams better understand their users.
Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau:

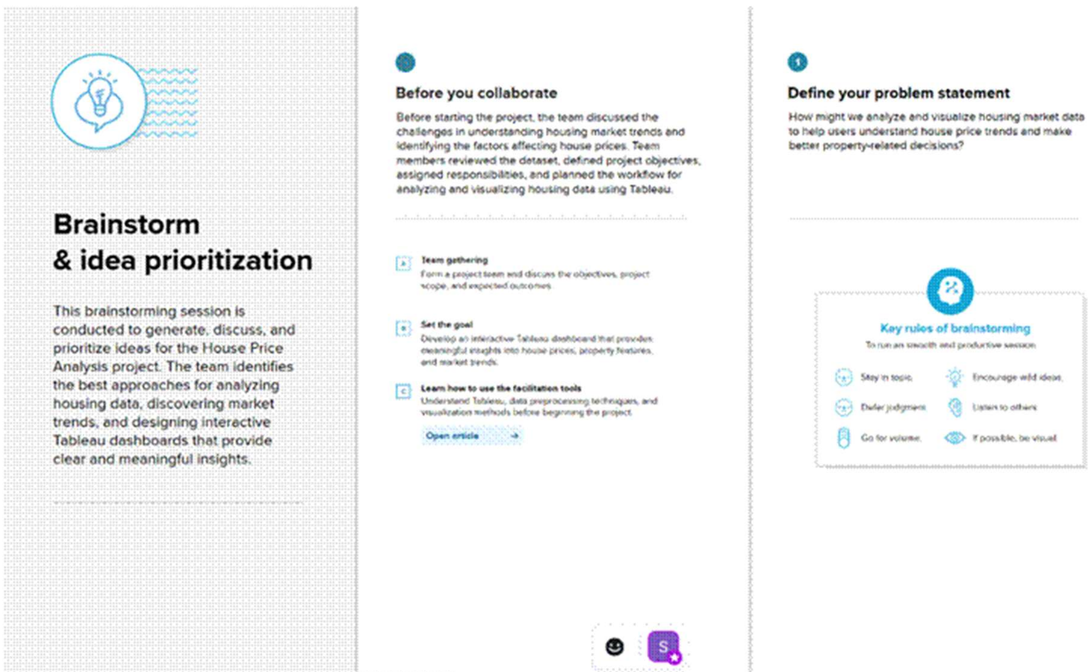


2.3 Brainstorming

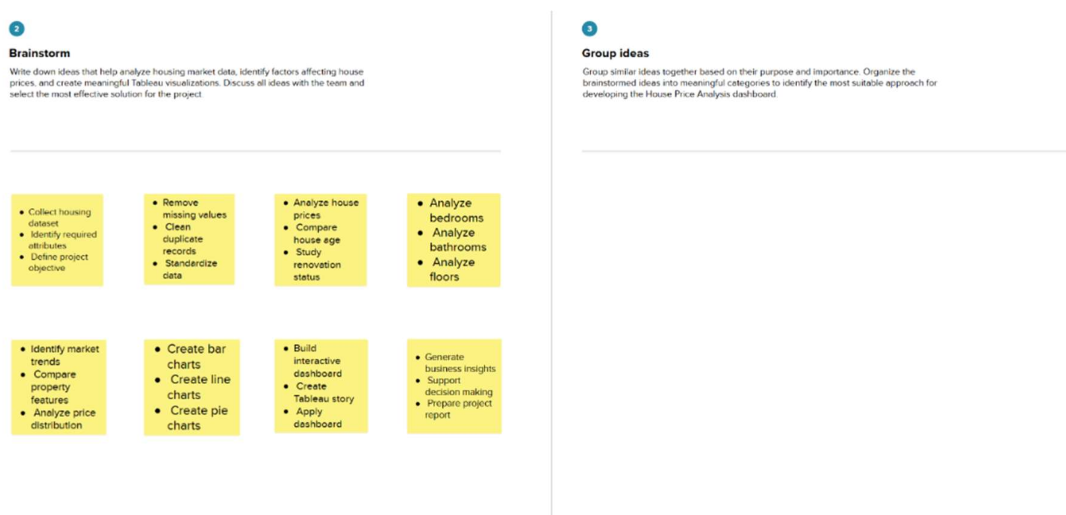
Brainstorm & Idea Prioritization :

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

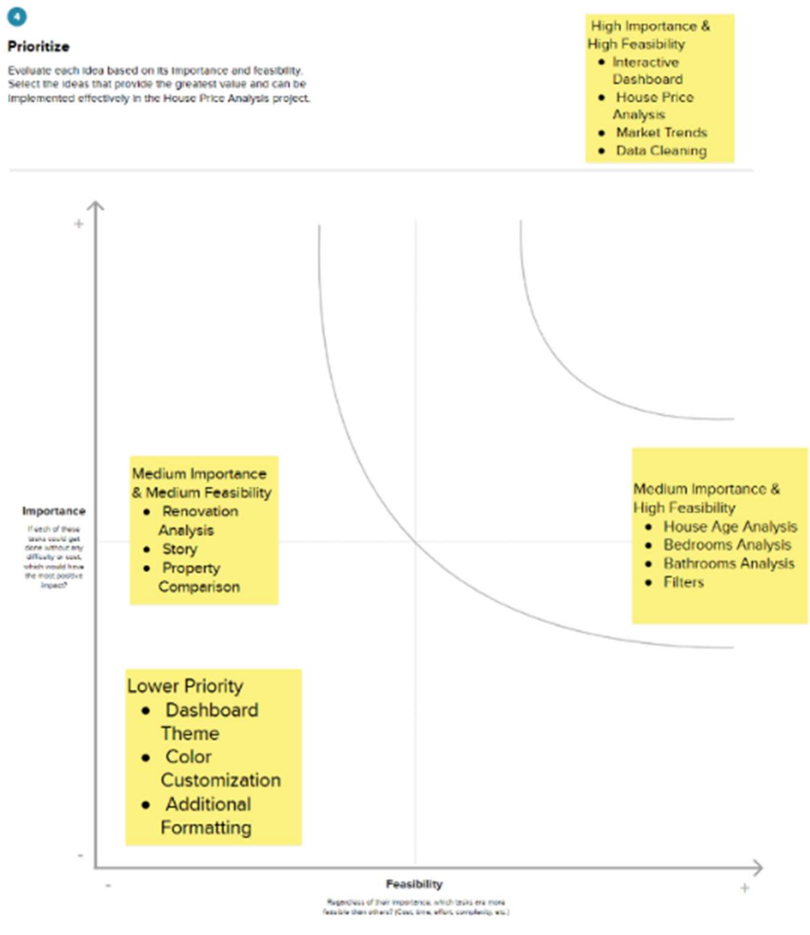
Step-1: Team Gathering, Collaboration and Select the Problem Statement



Step-2: Brainstorm, Idea Listing and Grouping

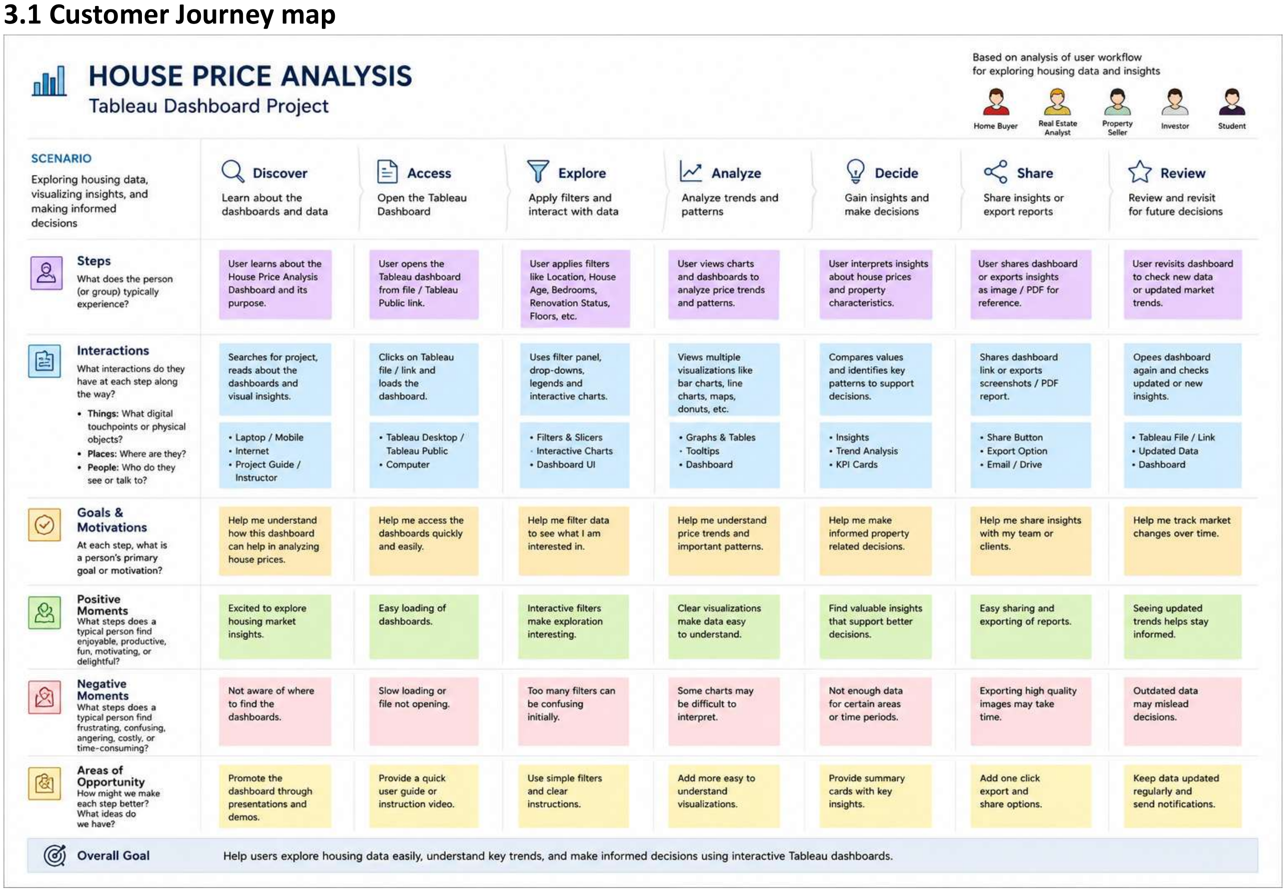


Step-3: Idea Prioritization



3. REQUIREMENT ANALYSIS

3.1 Customer Journey map



3.2 Solution Requirement

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection	Import housing dataset (CSV) into Tableau
		Verify dataset integrity and structure
FR-2	Data Preprocessing	Remove duplicate records
		Handle missing values and clean the dataset
FR-3	Data Visualization	Create charts and graphs for housing analysis
		Build interactive dashboards and Tableau Story
FR-4	Data Analysis	Analyze house prices and property features
		Generate business insights and trend analysis

Non-functional Requirements:

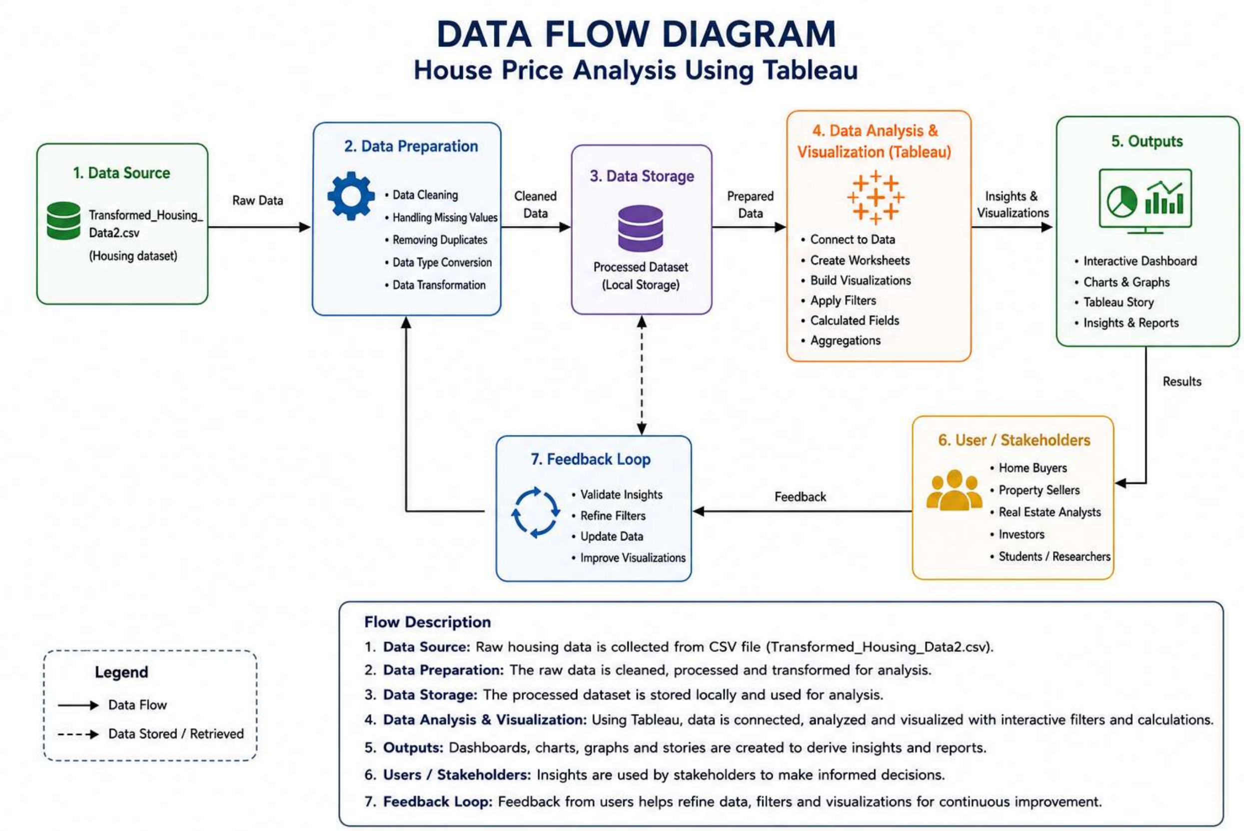
Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboards should be simple, interactive, and easy to understand.
NFR-2	Performance	Visualizations should load quickly and respond efficiently.
NFR-3	Reliability	The analysis should produce consistent and accurate results.
NFR-4	Data Accuracy	Housing data should be processed correctly without loss of information.
NFR-5	Scalability	The dashboard should support larger housing datasets in the future.
NFR-6	Maintainability	Dashboards should be easy to update when new housing data is available.

3.3 Data Flow Diagram

Data Flow Diagrams:

A Data Flow Diagram (DFD) illustrates how housing data moves through the House Price Analysis system. It represents the flow of data from the input dataset through preprocessing and analysis to the Tableau dashboards, where meaningful visualizations and business insights are generated. The DFD helps in understanding how data is processed and transformed into valuable information for users.



User Stories

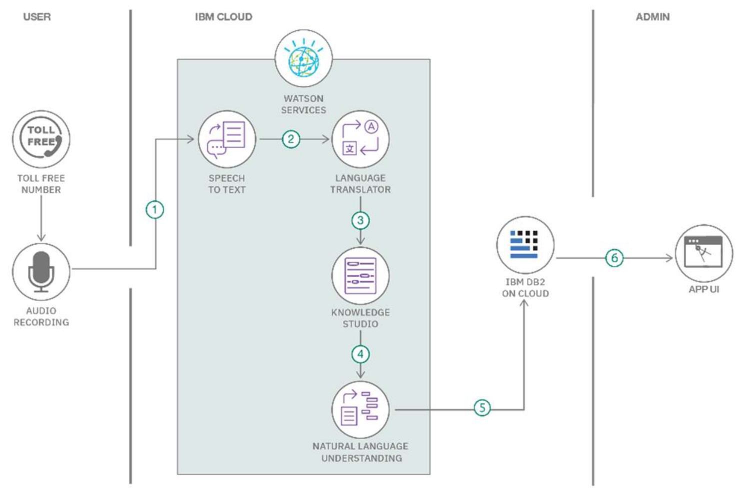
Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story No.	User Story / Task	Acceptance Criteria	Priority	Release	
User	Dataset Import	USN-1	As a user, I can import the housing dataset into Tableau.	Dataset is loaded successfully.	High	Sprint-1	
User	Data Cleaning	USN-2	As a user, I can clean and preprocess the housing data before analysis.	Clean dataset is ready for visualization.	High	Sprint-1	
User	Dashboard	USN-3	As a user, I can view interactive dashboards showing housing market trends.	Dashboards display accurate visualizations.	High	Sprint-1	
User	Filters	USN-4	As a user, I can apply filters to analyze specific housing information.	Filtered results are displayed correctly.	Medium	Sprint-2	
User	Analysis	USN-5	As a user, I can compare house prices based on different property features.	Comparative analysis is available.	High	Sprint-2	
User	Reports	USN-6	As a user, I can identify market trends and business insights from the dashboards.	Insights are presented clearly.	Medium	Sprint-2	

3.4 Technology Stack:

Technical Architecture:

The proposed **House Price Analysis** system is designed to analyze housing market data and present meaningful insights through interactive Tableau dashboards. The architecture consists of data collection, preprocessing, data analysis, visualization, and dashboard presentation. The processed housing data is transformed into interactive charts, dashboards, and stories that help users understand house price trends and property characteristics.



Guidelines:

- Import housing dataset (CSV).
- Perform data cleaning and preprocessing.
- Analyze housing market data.
- Create interactive visualizations.
- Build Tableau dashboards and story.
- Generate business

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	Data Source	Housing dataset in CSV format	CSV File
2	Data Preprocessing	Data cleaning and transformation	Tableau Prep / Tableau Desktop
3	Data Analysis	Analyze housing data and identify trends	Tableau Desktop
4	Data Visualization	Create interactive charts and dashboards	Tableau
5	Dashboard	Present business insights through dashboards	Tableau Dashboard
6	Story	Display analysis using Tableau Story	Tableau Story
7	File Storage	Store datasets and Tableau workbook	Local File System
8	Operating System	Platform used for development	Windows 10/11
9	Development Tool	Dashboard development environment	Tableau Desktop
10	Output	Interactive reports and visualizations	Tableau Public / Tableau Desktop

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Visualization Tool	Interactive dashboard development	Tableau Desktop
2	Data Accuracy	Accurate preprocessing and visualization	Tableau
3	Scalability	Supports additional housing data in future	Tableau
4	Availability	Dashboard can be accessed locally or through Tableau Public	Tableau Public
5	Performance	Fast filtering, visualization, and dashboard interaction	Tableau

4. PROJECT DESIGN

4.1 Problem Solution Fit:

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.

4.2 Proposed Solution

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1	**Problem Statement (Problem to be solved)**	Understanding housing market trends and identifying the factors that influence house prices is difficult when dealing with large and complex datasets. Users require an interactive and easy-to-understand visualization system to analyze housing information effectively.
2	**Idea / Solution Description**	Develop an interactive House Price Analysis dashboard using Tableau. The solution analyzes housing data, visualizes market trends, compares property characteristics, and presents meaningful insights through dashboards, charts, and stories to support informed decision-making.
3	**Novelty / Uniqueness**	The project combines multiple housing attributes such as house price, age, renovation status, bedrooms, bathrooms, floors, and sales trends into a single interactive Tableau dashboard, allowing users to perform detailed analysis using filters and visualizations.
4	**Social Impact / Customer Satisfaction**	The project helps home buyers, property sellers, and real estate analysts understand housing market trends more easily. Interactive visualizations simplify complex data, enabling users to make informed and confident property-related decisions.
5	**Business Model (Revenue Model)**	The dashboard can be used by real estate agencies, property consultants, and market analysts for housing market analysis. Organizations may use the dashboard as a decision-support tool or integrate it into their internal reporting systems.
6	**Scalability of the Solution**	The solution can be extended by incorporating larger housing datasets, additional property attributes, regional comparisons, and future housing market data to provide more comprehensive and detailed analysis.

4.3 Solution Architecture

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture of the House Price Analysis System:

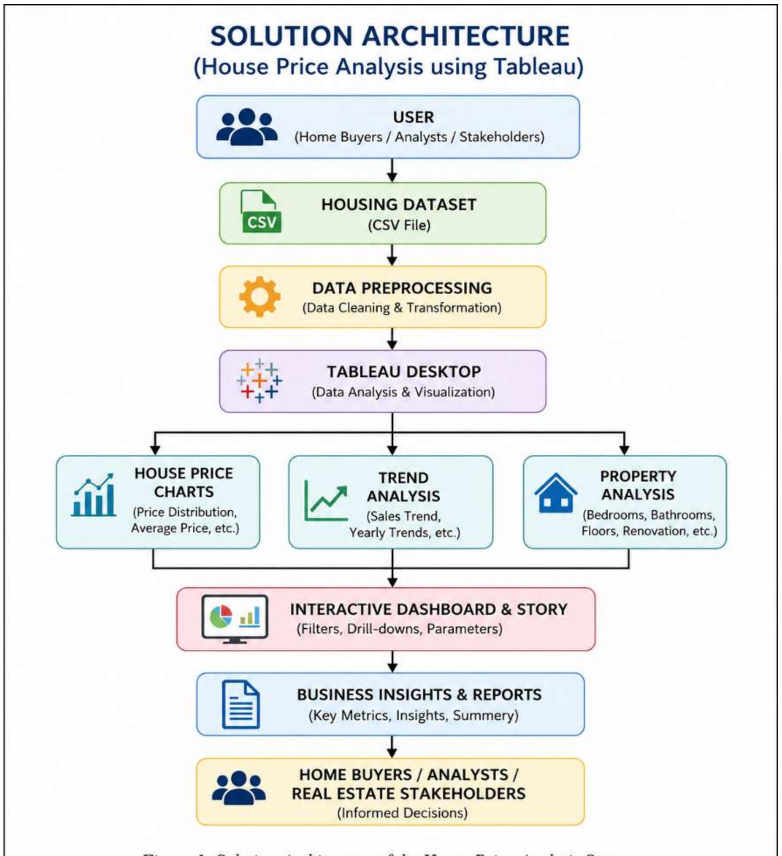


Figure 1: Solution Architecture of the House Price Analysis System

5. PROJECT PLANNING & SCHEDULING:

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story No.	User Story / Task	Story Points	Priority
Sprint-1	Data Collection	USN-1	Collect the housing dataset and verify its quality.	2	High
Sprint-1	Data Preprocessing	USN-2	Clean the dataset by removing duplicates and handling missing values.	3	High
Sprint-1	Data Transformation	USN-3	Organize and transform data for analysis in Tableau.	2	High
Sprint-2	Data Analysis	USN-4	Analyze house prices, house age, renovation status, bedrooms, bathrooms, and floors.	3	High
Sprint-2	Dashboard Design	USN-5	Create interactive charts and dashboards in Tableau.	5	High
Sprint-3	Story Creation	USN-6	Develop a Tableau Story to present business insights.	3	Medium
Sprint-3	Testing	USN-7	Validate dashboard accuracy and check visualizations.	2	Medium
Sprint-4	Documentation	USN-8	Prepare the final report and presentation.	2	High
Sprint-4	Deployment	USN-9	Publish the Tableau dashboard and complete project submission.	2	Medium
Sprint	**Functional Requirement (Epic)**	**User Story No.**	**User Story / Task**	**Story Points**	**Priority**

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	7	5 Days	01 Mar 2025	05 Mar 2025	7	05 Mar 2025
Sprint-2	8	5 Days	06 Mar 2025	10 Mar 2025	8	10 Mar 2025
Sprint-3	5	5 Days	11 Mar 2025	15 Mar 2025	5	15 Mar 2025
Sprint-4	4	5 Days	16 Mar 2025	20 Mar 2025	4	20 Mar 2025

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing:

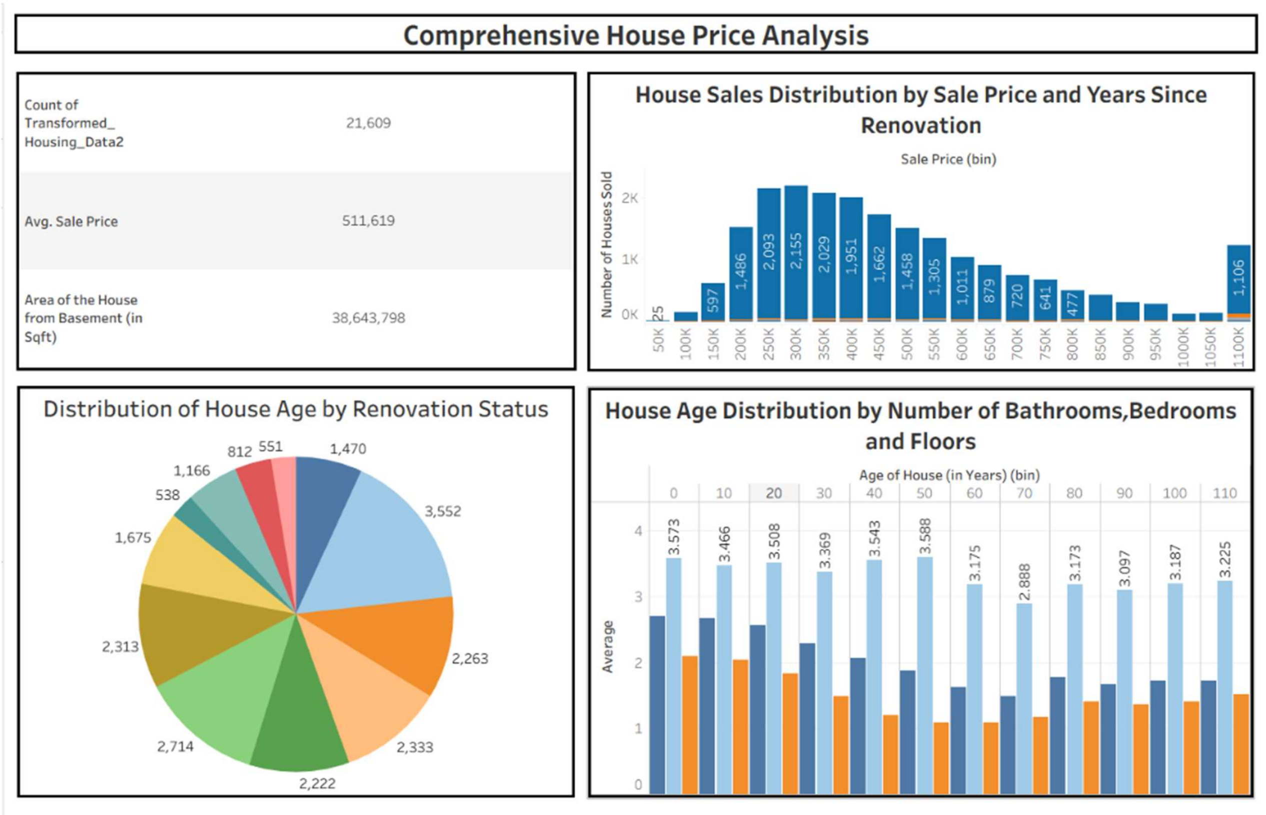
Model Performance Testing:

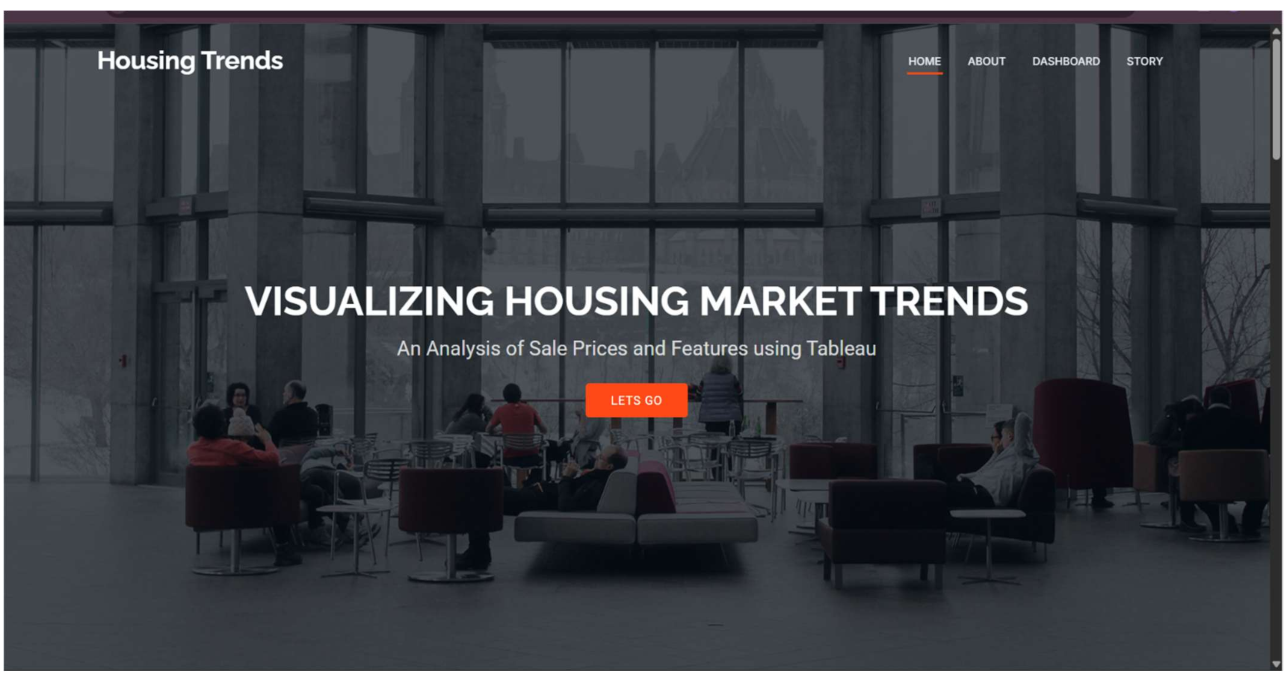
Project team shall fill the following information in model performance testing template.

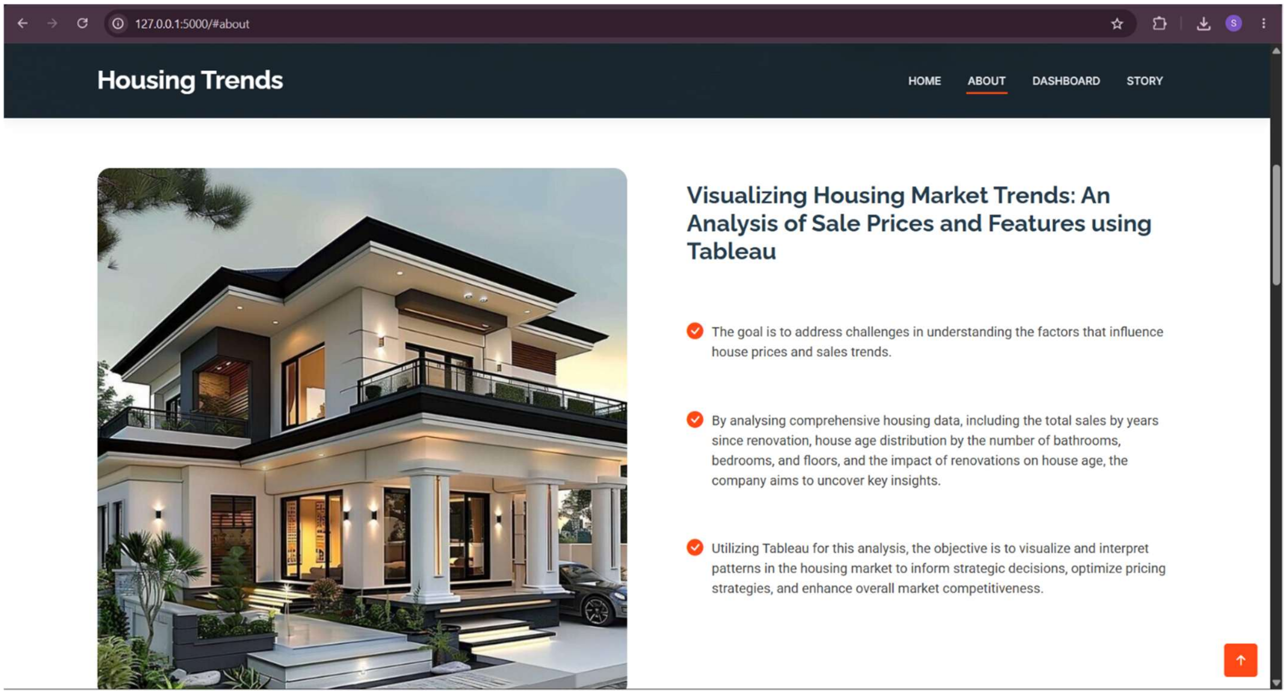
S.No.	Parameter	Values
1. **1**	**Data Rendered**	Housing dataset **Transformed_Housing_Data2.csv** was successfully imported into Tableau. All records and required fields were loaded correctly without data corruption.
2. **2**	**Data Preprocessing**	Missing values were handled, duplicate records were removed (where applicable), data types were verified, and the dataset was prepared for visualization in Tableau.
3	**Utilization of Filters**	Interactive filters were used to analyze housing data based on attributes such as **House Age, Renovation Status, Bedrooms, Bathrooms, Floors, and Years Since Renovation**, enabling dynamic exploration of the dataset.
4	**Calculated Fields Used**	Basic Tableau calculations and aggregations were used to summarize housing data for visualization. Calculated measures supported analysis of house price distribution and sales trends. *(If you did not create any custom calculated fields, you can write:* ***No custom calculated fields were created.****)*
5	**Dashboard Design**	**4 visualizations** were created to analyze different aspects of the housing dataset, including house age distribution, renovation status, property features, and total sales trends.
6	**Story Design**	**1 Tableau Story** was created to present the overall analysis by combining dashboard visualizations and explaining key insights related to house prices and housing market trends.

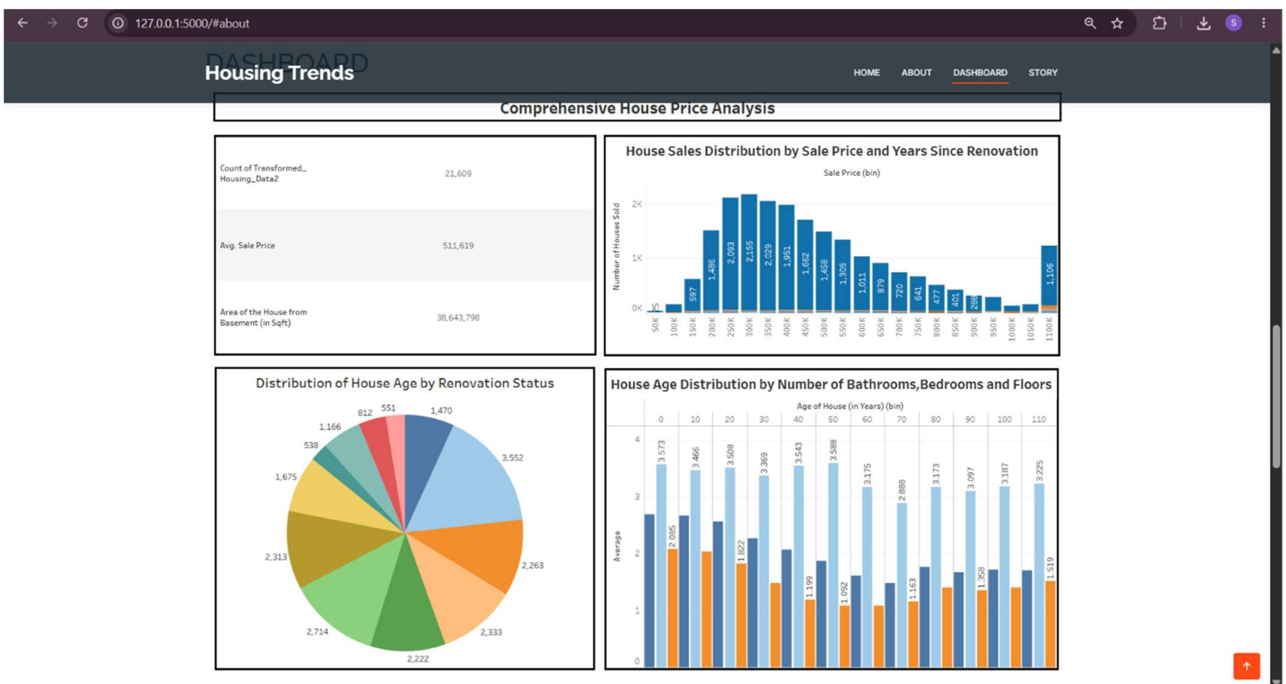
7. RESULTS

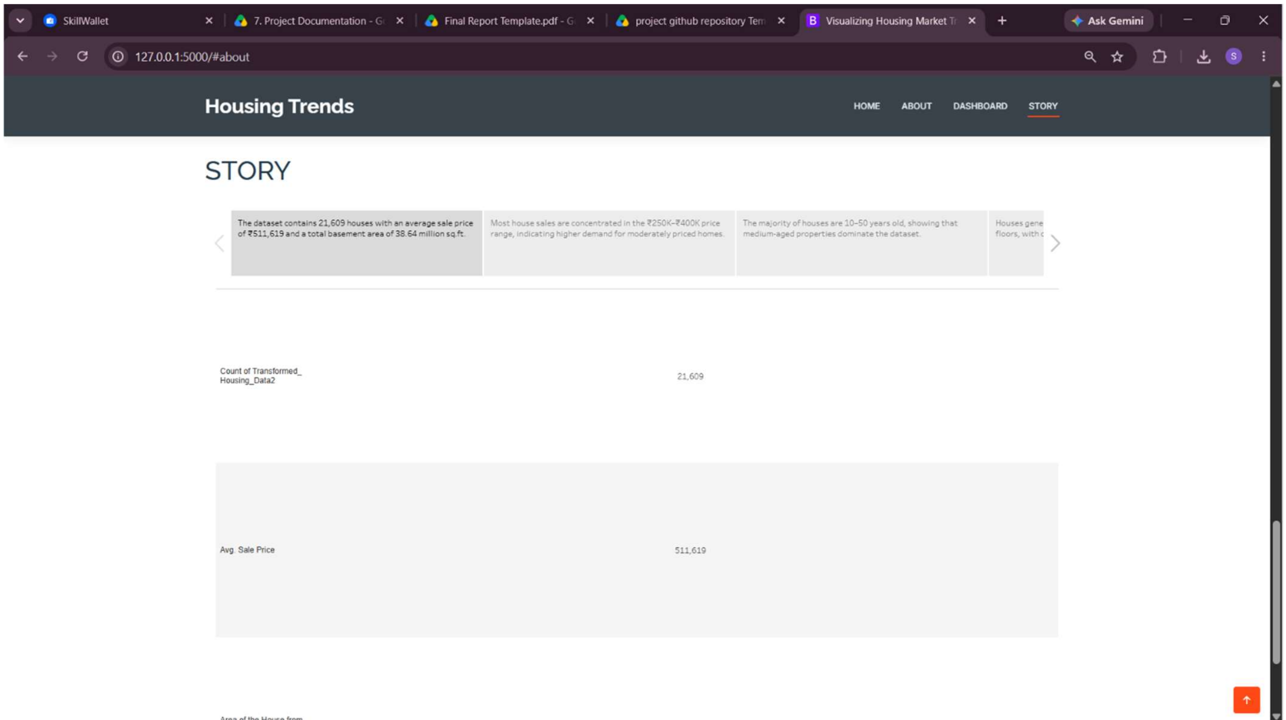
7.1 Output Screenshots






127.0.0.1:5000/#about
Housing Trends
HOME
ABOUT
DASHBOARD
STORY
Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
The goal is to address challenges in understanding the factors that influence house prices and sales trends.
By analysing comprehensive housing data, including the total sales by years since renovation, house age distribution by the number of bathrooms, bedrooms, and floors, and the impact of renovations on house age, the company aims to uncover key insights.
Utilizing Tableau for this analysis, the objective is to visualize and interpret patterns in the housing market to inform strategic decisions, optimize pricing strategies, and enhance overall market competitiveness.


127.0.0.1:5000/#about
Housing Trends
HOME
ABOUT
DASHBOARD
STORY
Comprehensive House Price Analysis
Count of Transformed_Housing_Data2
21,609
Avg. Sale Price
511,619
Area of the House from Basement (in Sqft)
38,643,798
House Sales Distribution by Sale Price and Years Since Renovation
Sale Price (bin)
Number of Houses Sold
Distribution of House Age by Renovation Status
House Age Distribution by Number of Bathrooms,Bedrooms and Floors
Age of House (in Years) (bin)
Average



8. ADVANTAGES & DISADVANTAGES

Advantages

- Interactive visualization of housing data.
- Easy comparison of property features.
- Better understanding of market trends.
- Supports informed decision-making.
- User-friendly Tableau dashboards.

Disadvantages

- Results depend on dataset quality.
- Limited to available housing data.
- Requires Tableau software for editing dashboards.
- Dashboard performance may decrease with very large datasets.

9. CONCLUSION

The House Price Analysis project successfully demonstrates how Tableau can transform raw housing data into meaningful visual insights. Through interactive dashboards, charts, and stories, the project enables users to analyze housing market trends, compare property characteristics, and understand factors influencing house prices. The developed solution provides an effective platform for exploring housing data and supports better data-driven decision-making.

10. FUTURE SCOPE

- Include additional housing datasets.
- Add regional and city-wise comparisons.
- Incorporate real-time housing market updates.
- Develop predictive price trend analysis.
- Enhance dashboard interactivity with more filters and visualizations.

11. APPENDIX:

Dataset Link

https://www.kaggle.com/datasets/rituparnaghosh18/transformed-housing-data-2

Tableau Workbook

https://drive.google.com/file/d/1pGKxqme9wOUixaN9BYYfPCzX_YWUH4q-/view?usp=sharing

Tableau Public Link

https://public.tableau.com/app/profile/shaik.nizaamuddin/viz/Book1_17826726804080/Dashboard1
https://public.tableau.com/app/profile/shaik.nizaamuddin/viz/Book2_17826729225890/Story1

GitHub Repository:

https://github.com/nizaam-07/Visualizing-Housing-Market-Trends-An-Analysis-of-Sale-Prices-and-Features-using-Tableau.git